# Modern Reinforcement Learning: A Practical Guide

## How to Use RL in 2025 with Gymnasium, PPO, and RLHF

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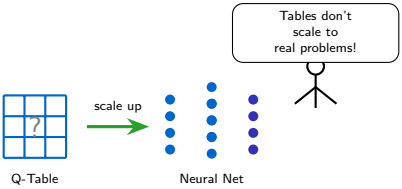
# You Know RL — Now How Do People Actually Use It?

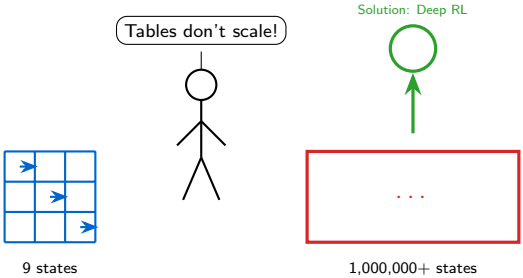



*You already know what RL is (L06b). Now: how do practitioners actually* **use** *it today?*

- Deep RL libraries that work out of the box
- PPO as the workhorse algorithm
- RLHF — how ChatGPT learned to be helpful

**From Q-tables to neural networks: modern RL scales to any problem.**

# The Scale Problem: Why Tables Break Down




Classic RL stores a table of values. Modern RL uses neural networks to handle any state space.

**A chess board has $\sim 10^{44}$ possible states. No table can store that. Neural networks generalize.**

# From Tables to Neural Networks

**Classic RL (L06b)**

- Q-table stores one value per state
- Works for small grids (9–100 states)
- Tabular lookup — no generalization

**Modern RL (Today)**

- Neural network approximates Q-values
- Works for any state space (millions+)
- Learns to generalize across states

## The Key Insight

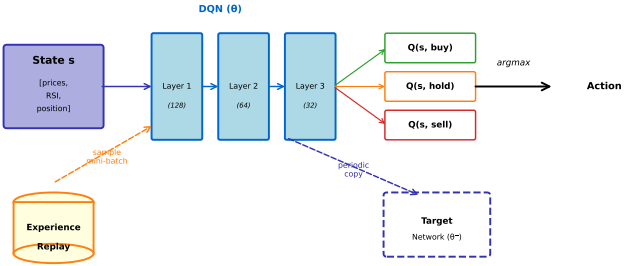
Replace the table with a neural network = Deep RL.

**DQN (2015) was the breakthrough: a neural network that plays Atari games from raw pixels.**

# DQN: The Neural Network That Plays Atari

***Loss Function***

$$L = (r + \gamma \max_{a'} Q(s', a'; \theta^{-}) - Q(s, a; \theta))^2$$




**Deep Q-Network Architecture for Trading**

- State (pixels, numbers) goes into the neural network
- Network outputs a Q-value for every possible action
- Pick the action with the highest Q-value

**DQN = Deep Q-Network. DeepMind used this to beat human experts at 49 Atari games (2015).**

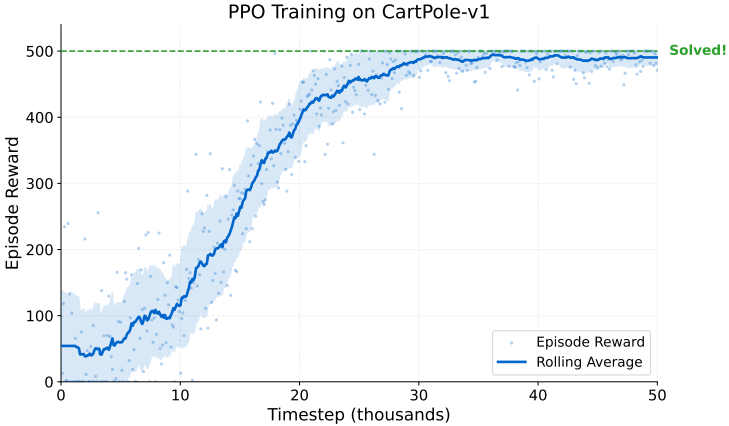
# The Modern RL Stack

| Layer | Tool |
|---|---|
| Environments | Gymnasium (successor to OpenAI Gym) |
| Algorithms | Stable-Baselines3 (PPO, SAC, DQN pre-built) |
| Training | GPU-accelerated (thousands of episodes/second) |
| Monitoring | Weights & Biases, TensorBoard |
| Deployment | ONNX export, sim-to-real transfer |

**Everything you need is open-source and pip-installable.**

**pip install gymnasium stable-baselines3. That is all you need to start training RL agents today.**

# PPO: The Workhorse Algorithm




- PPO learns fast: solves CartPole in ~30k timesteps
- Stable updates: no catastrophic performance drops
- The default choice for most RL tasks in 2025

**PPO = Proximal Policy Optimization (Schulman et al., 2017). Used by OpenAI, DeepMind, and most practitioners.**

# RLHF — How ChatGPT Learned to Be Helpful

1. **Train a base LLM** on text data (supervised learning)
2. **Humans rank outputs** — which response is better?
3. **RL (PPO!) fine-tunes** the LLM to match human preferences

**RLHF = Reinforcement Learning from Human Feedback**

The reward signal comes from human preferences, not a score function.

**The same PPO from Slide 7 is used to align ChatGPT.**

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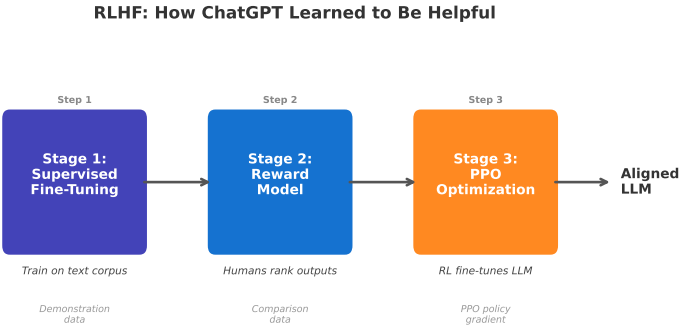
**InstructGPT paper (Ouyang et al., 2022) introduced RLHF for language models. Now standard for all frontier LLMs.**

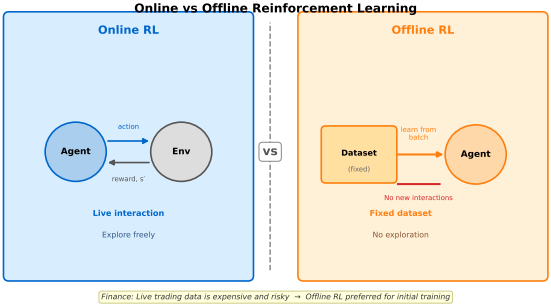
# What Does the RLHF Pipeline Look Like?




- Three stages: supervised learning, then human feedback, then PPO
- The reward model learns what humans prefer
- PPO optimizes the LLM to maximize the learned reward

**This exact 3-stage pipeline is used by OpenAI (ChatGPT), Anthropic (Claude), and Google (Gemini).**

# What If You Can't Explore? Offline RL




- **Online RL**: agent explores freely (games, robotics)
- **Offline RL**: learn from a fixed dataset — no new interaction
- Finance: live trading exploration is expensive and risky
- Key methods: CQL (Conservative Q-Learning), Decision Transformer

**Offline RL is the most finance-relevant RL frontier.**

**Offline RL learns from historical data only — critical when online exploration is too costly or dangerous.**

# FinRL: A Ready-Made Framework for Financial RL

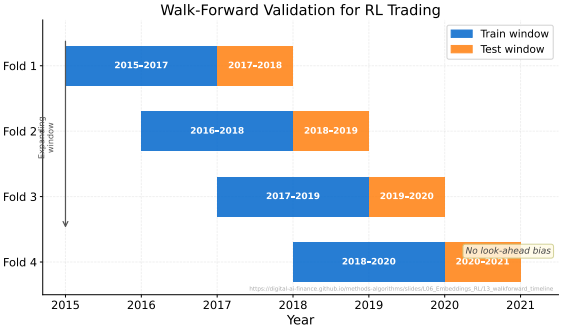
```
from finrl.agents.stablebaselines3 import DRLAgent
from finrl.meta.env_stock_trading import StockTradingEnv

env = StockTradingEnv(df=stock_data)
agent = DRLAgent(env)
model = agent.get_model("ppo")
trained = agent.train_model(model, total_timesteps=50000)
```

- Pre-built trading environments (stocks, crypto, portfolio)
- Handles data download, feature engineering, backtesting
- Supports PPO, A2C, DDPG, SAC out of the box

**FinRL: github.com/AI4Finance-Foundation/FinRL (Liu et al., 2021). pip install finrl.**

# Where Modern RL Shines




- **Game AI**: AlphaGo, OpenAI Five
- **Robotics**: sim-to-real transfer
- **Finance**: walk-forward validated trading

Train millions of episodes in simulation (free, fast, safe), then deploy.

**Sim-to-real: train in simulation where mistakes are free, then fine-tune on real data.**

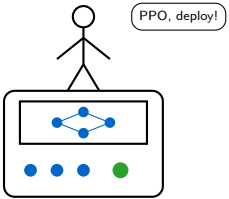
# Try It Yourself — Five Lines of Python

```
from stable_baselines3 import PPO
import gymnasium as gym
env = gym.make("CartPole-v1")
model = PPO("MlpPolicy", env)
model.learn(total_timesteps=50_000)
```

**Five lines. That is a complete RL training pipeline.**

**pip install stable-baselines3[extra]. CartPole is the "Hello World" of RL. Try LunarLander-v3 next!**

# Key Takeaways




1. Modern RL uses neural networks, not tables (Deep RL)
2. PPO is the workhorse algorithm; RLHF uses it to align LLMs
3. Gymnasium + Stable-Baselines3 = 5 lines to a trained agent

**Next: try the RL notebook to train your own PPO agent!**